

Retail Sentiment, Valuation Regimes, and the Cross-Section of European Equity Returns:

A Fama–French 25-Portfolio Analysis of the STOXX Universe

Corrected and revised edition

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Abstract

European retail equity participation roughly doubled between 2018 and 2022. Trade Republic grew from launch to four million users in three years; DEGIRO processed approximately ten times its pre-2019 order volume; Trading 212 suspended new UK account openings in January 2021 due to demand. The standard Fama–French five-factor model was built on data from before any of this existed. This paper asks whether the structural shift left a detectable footprint in the cross-section of European equity returns.

We propose the *Retail Sentiment Factor* (RSF) and test it across the STOXX universe using a Fama–French 25-portfolio double-sort on (i) trailing price-to-earnings ratio (PE) and (ii) *Signed Abnormal Turnover* (SAT), a retail sentiment proxy grounded in [Baker and Stein \(2004\)](#). Under a strictly enforced sort-at- t / hold-at- $t+1$ convention, we find *no evidence* that SAT predicts next-month returns: SAT quintile spreads lie between -0.15% and $+0.24\%$ per month with $|t| < 0.7$ throughout, FF6 alphas are small and almost uniformly insignificant, the PE–SAT interaction Δ_{Cross} is indistinguishable from zero, and Fama–MacBeth slopes carry no significance. Daily-horizon variants of the test reach the same conclusion.

The paper also documents a methodological trap specific to signed-volume sentiment measures. Because SAT embeds the sign of the formation-month return, pairing a month- t sort with month- t returns produces spectacular but entirely mechanical results — SAT “spreads” of 8–9% per month with Newey–West t -statistics above 19. An earlier draft of this paper reported exactly those numbers. We present the contaminated and corrected results side by side as a cautionary diagnostic for the retail-sentiment literature.

The null predictive result does not make retail sentiment irrelevant to asset pricing. A companion paper constructs RSF as a Fama–French-style 2×3 size/SAT factor and shows it is *not spanned* by the FF5 + momentum factors (spanning alpha 0.22%/month, $t = 2.24$, $R^2 = 0.03$): retail sentiment is better understood as a distinct source of common return variation to be controlled for, rather than a return forecaster.

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1 Introduction

Between 2019 and 2022, the retail share of European equity order flow approximately doubled, rising from around 10–12% to an estimated 20–25% (Debbaut et al., 2023). Three platform-level figures illustrate the pace: Trade Republic grew from launch to four million users in three years; DEGIRO processed roughly ten times its pre-2019 order volume during the 2020–2021 period; Trading 212 suspended new UK account openings in January 2021 due to demand it could not process. This is not a gradual drift. It is a step change in the composition of market participants, concentrated in a narrow window, with a plausible causal mechanism: the simultaneous arrival of zero-commission trading, fractional shares, and social-media-coordinated investment communities.

The Fama–French five-factor model (Fama and French, 1993, 2015) was identified on data from an era when European retail participation was thin and fragmented. The question this paper addresses is whether the post-2020 retail surge produced an *additional* systematic cross-sectional pattern that the existing model does not capture — specifically, whether a valuation-conditional retail sentiment measure predicts next-month returns.

We propose the **Retail Sentiment Factor (RSF)**. The core mechanism is that retail sentiment does not affect all stocks equally: its impact is modulated by valuation. When retail enthusiasm surges, it concentrates in high-PE growth stocks where narratives are active and extrapolative expectations dominate. When sentiment falls, it is low-PE value stocks already labelled “cheap for a reason” that absorb the most intense selling.

This corrected edition of the paper reports two findings. First, the *predictive* version of the hypothesis fails: under a strictly enforced sort-at- t / hold-at- $t+1$ convention, SAT does not forecast next-month returns at any horizon we examine, in portfolio sorts, Fama–MacBeth regressions, or daily variants. Second, the paper documents *why* an earlier draft reported the opposite. Signed-volume sentiment measures are unusually vulnerable to look-ahead bias: SAT is signed by the formation-month return, so any pipeline that pairs the month- t sort with month- t returns manufactures a mechanical 8–9%-per-month “spread” with t -statistics above 19. Section 9 presents the two sets of results side by side. We believe this diagnostic is of independent value to the retail-sentiment literature, where signed measures (order imbalance, signed volume, signed turnover) are standard building blocks.

The Measurement Challenge and Our Solution

Testing the RSF in a European context requires solving a measurement problem that the US literature does not face. The Boehmer et al. (2021) algorithm (BJZZ) identifies retail-initiated trades in NYSE TAQ data by detecting subpenny price improvements from wholesale internalisation, a feature of Regulation NMS and the US payment-for-order-flow ecosystem. The EU’s MiFIR framework never permitted the same internalisation

structure, and Regulation (EU) 2024/791 has now banned PFOF outright (MiFIR, 2023). There is no European TAQ equivalent.

We develop instead **Signed Abnormal Turnover (SAT)**, grounded in Baker and Stein (2004). Their central result is that short-selling friction creates a systematic asymmetry: retail investors, who rarely short, disproportionately drive exchange-reported volume when they are optimistic and buying. Multiplying abnormal turnover by the sign of the contemporaneous return therefore separates the retail signal from institutional noise. SAT requires only daily price, volume, and shares outstanding — standard fields available for every STOXX constituent.

Contributions

This paper makes four contributions. First, SAT provides a theoretically grounded, practically implementable retail sentiment proxy for European equity markets where BJZZ identification is unavailable. Second, this is the first systematic test of valuation-conditional retail sentiment effects in the STOXX universe — and the answer, under clean alignment, is a null. Third, the look-ahead diagnostic of Section 9 quantifies exactly how large the spurious effect from contemporaneous sort–return pairing is for signed-volume measures, providing a concrete benchmark for referees and replicators. Fourth, the companion factor-spanning analysis (summarised in Section 10) shows the constructive reinterpretation: RSF is unspanned by FF5+momentum and belongs on the right-hand side of factor regressions, not in a forecasting rule.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 documents the 2020 regime shift. Section 4 develops the formal hypotheses. Section 5 describes the data. Section 6 details portfolio construction and the alignment convention. Section 7 presents the econometric framework. Section 8 reports the corrected results. Section 9 presents the look-ahead diagnostic. Section 10 summarises the companion spanning analysis. Section 11 concludes.

2 Related Literature

2.1 Sentiment, Turnover, and the Cross-Section

Baker and Wurgler (2006, 2007) establish the empirical foundation: a composite sentiment index predicts the cross-section of returns, with stronger effects among hard-to-value, difficult-to-arbitrage stocks. Baker and Stein (2004) provide the theoretical mechanism directly underlying our SAT measure: if irrational investors are more likely to speculate when optimistic than pessimistic (due to short-selling frictions), market liquidity and turnover are higher in bull markets, and high turnover is a direct signal of sentiment-driven retail activity.

The turnover-as-sentiment literature has a long history. [Lee et al. \(1991\)](#) use turnover to identify noise trader activity. [Statman et al. \(2006\)](#) show that individual investors trade more after market gains, consistent with overconfidence and extrapolative expectations. Most directly, [Kaniel et al. \(2008\)](#) show that individual investor order imbalance predicts short-term returns. Our SAT measure is the signed, abnormal version of this imbalance.

2.2 Retail Investor Behaviour: US Evidence and the European Gap

The post-2020 retail literature is largely US-focused. [Boehmer et al. \(2021\)](#) introduce the BJZZ algorithm and document that retail order imbalance significantly predicts next-day returns. [Welch \(2022\)](#) shows that aggregate retail imbalance from Robinhood predicts short-term momentum. [Hasso et al. \(2022\)](#) document herding behaviour in meme-stock episodes. European retail investor behaviour has received less systematic attention. [Debaut et al. \(2023\)](#) document rising retail equity ownership across the EU post-2020 but do not study cross-sectional return implications. We fill this gap.

2.3 Momentum and Compressed Horizons

Classical momentum ([Jegadeesh and Titman, 1993](#)) operates on 3–12 month formation windows. [Da et al. \(2011\)](#) show that retail attention predicts short-term return continuation at weekly horizons. [Han et al. \(2022\)](#) document that social media sentiment predicts returns at daily-to-weekly frequencies. Our SAT measure operates at monthly frequency but is constructed from daily data; Section 8.6 additionally tests daily and weekly variants.

2.4 Valuation and Retail Attention

[Barber and Odean \(2008\)](#) show retail investors are disproportionately buyers of attention-grabbing stocks. [Fang and Peress \(2009\)](#) document that media coverage reduces information asymmetry. The PE interaction we propose extends these findings: retail attention is systematically higher for growth stocks with active narratives (high PE) and lower for value stocks perceived as “broken” (low PE).

2.5 The Factor Zoo

[Harvey et al. \(2016\)](#) set the appropriate t -statistic threshold for a new factor at approximately 3.0, having catalogued 316 factors with a median t -statistic of 4.0. Their diagnosis is that most published factors are the product of data mining. This paper adds a complementary diagnosis: some spectacular factor candidates are the product not of mining

but of *mechanical contamination*, where the sorting variable embeds the contemporaneous return. The look-ahead comparison in Section 9 shows that such contamination can dwarf any plausible data-mining effect — t -statistics near 30 from a signal with no genuine predictive content.

3 The 2020 Regime Shift in European Retail Participation

The RSF mechanism requires a critical mass of retail participation to generate detectable cross-sectional patterns. Before that mass existed, the factor should be empirically absent.

3.1 Evidence for the Structural Break

Table 1 summarises the key indicators of the participation shift.

Table 1: Indicators of the Structural Shift in European Retail Equity Participation

Indicator	Pre-2020	Post-2020
Retail share of EU equity order flow	~10–12%	~20–25% (Debbaut et al., 2023)
Zero-commission platforms (Europe)	DEGIRO only (founded 2013)	Trade Republic, Scalable Capital, Trading 212, Revolut Invest all active
Trade Republic user base	<0.5m at end-2019	>4m by end-2022
Fractional share availability	Rare; DEGIRO partial	Standard offering across major platforms
Reddit WallStreetBets membership	~1m (Jan 2020)	>13m (Feb 2021); European equivalents followed

3.2 The Regime Break Date and Test

We set the regime break at **31 December 2019**. This date precedes the COVID-19 market events of March 2020, coincides with Trade Republic’s commercial launch, and is *predetermined* — selected before any subsample results were examined. The primary test is a Chow-type structural break regression on the monthly Fama–MacBeth interaction slope:

$$\hat{\lambda}_{\text{Int},t} = \alpha + \beta \cdot \mathbf{1}[t > \text{Dec 2019}] + \varepsilon_t \quad (1)$$

with Newey–West standard errors (6 lags). A positive and significant $\hat{\beta}$ would constitute direct evidence that the PE–SAT interaction strengthened with the retail surge.

4 Theoretical Framework and Hypotheses

4.1 The Core Mechanism

Three mutually reinforcing mechanisms could generate an RSF pattern.

Short-selling friction and retail asymmetry. Baker and Stein (2004) establish that when short-selling is costly — which it is for retail investors — irrational optimists are more likely to trade than irrational pessimists. Turnover spikes are therefore disproportionately retail-buy-driven.

Narrative-dependent extrapolation. Shiller (2020) argues that investors process information through narratives. A high-PE stock embodies a growth narrative that retail investors amplify through extrapolative expectations (Barberis, 2018); a low-PE stock carries “value trap” narratives that trigger avoidance during bullish episodes and amplified selling during bearish ones.

Limits to arbitrage. Shleifer and Vishny (1997) establish that institutional arbitrageurs face constraints preventing them from fully correcting retail-driven mispricing, potentially allowing an RSF pattern to persist at portfolio level.

4.2 Formal Hypotheses

Hypothesis 1 (Growth Amplification). *Within bullish SAT quintiles ($s \in \{4, 5\}$), mean next-month excess returns are monotonically increasing across PE quintiles.*

Hypothesis 2 (Value-Trap Penalty). *Within bearish SAT quintiles ($s \in \{1, 2\}$), mean next-month excess returns are monotonically increasing across PE quintiles, and the PE spread is larger for bearish quintiles than bullish ones.*

Hypothesis 3 (Sentiment Non-Linearity). *The SAT return premium within each PE quintile is asymmetric between the extreme SAT quintiles (the sentiment smile).*

Hypothesis 4 (Post-2020 Regime Amplification). *The PE–SAT interaction is approximately zero pre-2020 and positive and significant post-2020, with $\hat{\beta} > 0$ in the Chow regression (1).*

5 Data

5.1 Data Sources, Universe, and Period

Daily total returns, volume, shares outstanding, market capitalisation, trailing PE (TR.PE), and GICS sector codes are sourced from Refinitiv Eikon for the STOXX Europe 600 con-

stituent universe. European Fama–French five-factor and momentum returns and the risk-free rate come from the Ken French Data Library (Europe daily files, aggregated to monthly). Financial firms (GICS sector 40) are excluded. The sample runs from June 2009 to December 2025; after the 252-day AR warm-up and eligibility screens, the estimation panel covers **187 months, 478 stocks, and 69,492 stock-month observations**. All returns are EUR-denominated Refinitiv total returns (adjusted for splits and dividends).

Stocks must have at least 6 valid observations in the preceding 8 months to be eligible for a given month’s sort (a rolling, strictly point-in-time requirement: new entrants earn eligibility as history accumulates; stocks that exit and re-enter must re-earn it).

5.2 The Signed Abnormal Turnover (SAT) Proxy

Definition 1 (Turnover and Abnormal Turnover). *Daily turnover is $\text{Turn}_{i,d} = \text{Volume}_{i,d} / \text{SharesOut}_{i,d}$. Abnormal turnover is the out-of-sample residual from an AR(2) model of log turnover estimated on a rolling 252-day window ending the day before d :*

$$\text{ATV}_{i,d} = \exp(\hat{\varepsilon}_{i,d}) - 1. \quad (2)$$

Definition 2 (Signed Abnormal Turnover). *Daily SAT is $\text{SAT}_{i,d} = \text{ATV}_{i,d} \times \text{sign}(r_{i,d})$; zero-return days are excluded. Monthly SAT is the mean of daily SAT over the month (minimum 10 valid days), winsorised at 1%/99% and z-scored cross-sectionally within each month.*

Remark 1 (The sign convention is the look-ahead hazard). *Because $\text{SAT}_{i,t}$ is an average of $\text{ATV} \times \text{sign}(r)$ over month t , a stock’s month- t SAT is mechanically positively correlated with its month- t return. Any research design that measures portfolio returns over the same month used for the SAT sort will therefore “discover” an enormous spurious SAT premium. Section 9 quantifies this.*

5.3 The PE Variable

Trailing PE is winsorised at the 1st/99th percentile cross-sectionally within each month; negative-PE observations are excluded. Quintile breakpoints use the full eligible STOXX cross-section.

6 Portfolio Construction

6.1 The FF-25 Double Sort and the Alignment Convention

At the end of each formation month t , all eligible stocks are independently sorted into quintiles on trailing PE and monthly SAT, both computed from month- t data only. The

25 intersection portfolios are held over month $t+1$:

$$r_{k,s,t+1} = \sum_{i \in \mathcal{P}_t(k,s)} w_{i,t} r_{i,t+1}, \quad w_{i,t} = \frac{\text{ME}_{i,t}}{\sum_{j \in \mathcal{P}_t(k,s)} \text{ME}_{j,t}}, \quad (3)$$

with value weights fixed at month- t market equity. Portfolios are rebalanced monthly.

Remark 2 (Enforcement). *The alignment is enforced in a single shared code path used by every monthly test in the paper (portfolio sorts, Fama–MacBeth, parametric analysis). The forward return of stock i at formation month t is defined only if the stock’s next observation is exactly month $t+1$; across listing gaps and suspensions the forward return is set to missing rather than silently taking the next available observation. Automated tests verify that a panel in which the sorting variable drives only formation-month returns produces exactly zero portfolio returns.*

6.2 Spread Statistics

Portfolio (k, s) has PE quintile k ($1 = \text{lowest}$) and SAT quintile s ($1 = \text{most bearish}$). The three primary statistics are

$$\Delta_{\text{PE}}(s) = \bar{r}_{5,s} - \bar{r}_{1,s}, \quad (4)$$

$$\Delta_{\text{SAT}}(k) = \bar{r}_{k,5} - \bar{r}_{k,1}, \quad (5)$$

$$\Delta_{\text{Cross}} = \Delta_{\text{PE}}(5) - \Delta_{\text{PE}}(1). \quad (6)$$

7 Econometric Framework

For each of the 25 portfolios we estimate FF6 time-series alphas,

$$r_{k,s,t} - r_{f,t} = \alpha_{k,s} + \beta'_{k,s} \mathbf{F}_t + \varepsilon_{k,s,t}, \quad (7)$$

where $\mathbf{F}_t$ collects Mkt-RF, SMB, HML, RMW, CMA, and UMD, using Newey–West standard errors (6 lags). Stock-level predictability is tested with monthly Fama–MacBeth regressions

$$r_{i,t+1} - r_{f,t+1} = \lambda_0 + \lambda_{\text{PE}} \text{PEQ}_{i,t} + \lambda_{\text{SAT}} \text{SATQ}_{i,t} + \lambda_{\text{Int}} (\text{PEQ}_{i,t} \times \text{SATQ}_{i,t}) + \varepsilon_{i,t+1}, \quad (8)$$

using quintile ranks (robust to the heavy tails of raw SAT), with forward excess returns winsorised at 1%/99% within each month and Newey–West (6 lags) on the time series of monthly slopes. The parametric complement estimates the SAT premium within each PE quintile and fits a Weibull $f(\text{PE})$ curve to the five $(\text{PE}_k^{\text{med}}, \hat{\lambda}_{\text{SAT}}^{(k)})$ pairs.

8 Results

8.1 Mean Monthly Excess Returns: The 5×5 Matrix

Table 2 reports mean monthly excess returns for the 25 portfolios, with Newey–West t -statistics on the spread statistics.

Table 2: Mean Monthly Excess Returns: FF-25 Double Sort on PE and SAT (Corrected Alignment)

PE Quintile	SAT Quintile					Q5–Q1 (t)
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)	
Q1 (Low PE)	0.77%	0.92%	0.58%	0.84%	1.01%	0.24% (0.65)
Q2	0.66%	1.09%	0.86%	0.90%	0.84%	0.18% (0.65)
Q3	1.02%	0.30%	0.93%	0.70%	0.87%	-0.15% (-0.54)
Q4	1.13%	0.89%	0.46%	1.35%	1.07%	-0.06% (-0.21)
Q5 (High PE)	0.65%	1.17%	0.72%	1.01%	0.84%	0.20% (0.59)
Q5–Q1	-0.12%	0.25%	0.11%	0.16%	-0.16%	-0.05% (-0.10)
(t)	(-0.26)	(0.66)	(0.30)	(0.40)	(-0.44)	

Notes: Value-weighted mean monthly excess returns for 25 portfolios double-sorted at the end of formation month t on trailing PE quintile (rows) and monthly SAT quintile (columns), held over month $t+1$. Sample: June 2009 – December 2025 (187 formation months, 478 stocks, 69,492 stock-months). Financials excluded. t -statistics (Newey–West, 6 lags) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The corrected matrix contains none of the structure the RSF mechanism predicts. The SAT spreads $\Delta_{\text{SAT}}(k)$ range from -0.15% to $+0.24\%$ per month across PE quintiles, with every $|t| < 0.7$. The PE spreads are similarly small and insignificant. The interaction statistic Δ_{Cross} is -0.05% per month ($t = -0.10$): zero for all practical purposes. Hypotheses 1–3 are not supported — no monotone gradient, no corner concentration, no smile.

8.2 FF6 Time-Series Alphas

Table 3: FF6 Monthly Alphas: 25 Portfolios Double-Sorted on PE and SAT (Corrected Alignment)

PE Quintile	SAT Quintile				
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)
Q1 (Low PE)	0.20%	0.32%	0.09%	0.47%	0.67%*
Q2	0.16%	0.44%	0.39%	0.46%	0.34%
Q3	0.65%**	-0.19%	0.30%	0.05%	0.51%*
Q4	0.43%	0.13%	0.04%	0.61%**	0.42%
Q5 (High PE)	0.31%	0.48%*	0.15%	0.44%*	0.17%

Notes: FF6 monthly alphas from time-series regressions of holding-month portfolio excess returns on Mkt-RF, SMB, HML, RMW, CMA, and UMD. Newey–West standard errors, 6 lags. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Alphas lie between -0.19% and $+0.67\%$ per month; only 2 of 25 exceed $|t| = 2$, which is approximately what one expects from 25 tests under the null. There is no SAT gradient in the alpha matrix. The 8–9%-per-month alpha structure reported in the earlier draft (Section 9) does not survive correct alignment.

8.3 Fama–MacBeth Regressions

Table 4: Fama–MacBeth Cross-Sectional Regressions

Variable	$\hat{\lambda}$	t -stat	p -value	N (months)
const	0.011327	3.64	0.000	186
PEQ	-0.000278	-0.45	0.654	186
SATQ	0.000252	0.63	0.527	186
PEQ×SATQ	-0.000074	-0.63	0.531	186

Notes: Time-series means of monthly cross-sectional slopes from (8). Dependent variable: one-month-ahead excess return. PEQ, SATQ are quintile ranks. Newey–West standard errors, 6 lags.

All three slopes are insignificant ($|t| < 0.7$). In the earlier draft the insignificant FMB slopes stood in “apparent tension” with the enormous portfolio spreads and required an elaborate reconciliation; under correct alignment the tension disappears because both methods agree: there is no predictive relationship. The FMB regressions were the one

component of the original pipeline that used properly lagged returns — their nulls were the early warning.

8.4 Parametric PE–SAT Interaction

Table 5: PE-Quintile-Conditional SAT Premium: Parametric Analysis

PE Quintile	Med. PE	Full Sample		Pre-2020		Post-2020	
		$\hat{\lambda}$	t	$\hat{\lambda}$	t	$\hat{\lambda}$	t
Q1 (Low)	8.0	0.00003	0.07	0.00007	0.14	-0.00004	-0.06
Q2	13.3	0.00063	1.79	0.00041	0.92	0.00101	1.49
Q3	18.7	-0.00035	-0.86	-0.00008	-0.17	-0.00079	-1.02
Q4	25.4	-0.00032	-0.92	-0.00034	-0.75	-0.00029	-0.53
Q5 (High)	42.9	0.00012	0.25	0.00027	0.50	-0.00013	-0.14
<i>Weibull $f(\text{PE})$ fit diagnostics</i>							
R^2		-0.002		0.431		0.580	
peak PE		10.7		10.8		10.6	
scale		0.0012		0.0059		0.0028	
asymmetry		1.36		2.78		1.40	

Notes: For each PE quintile, monthly Fama–MacBeth cross-sections on the quintile subsample with SATQ as sole regressor; $\hat{\lambda}$ is the Newey–West time-series mean. Regime split at March 2020. Weibull $f(\text{PE})$ estimated by NLS on the five $(\text{PE}^{\text{med}}, \hat{\lambda})$ pairs.

No within-quintile SAT premium is statistically distinguishable from zero in any regime ($|t| < 1.8$ throughout). The Weibull fit diagnostics should be read with corresponding scepticism: with five noisy points and four parameters, subsample R^2 values are not informative about structure. We retain the table for completeness but draw no structural conclusion from it.

8.5 Regime Analysis

Table 6: Chow-Type Test for a Post-2019 Shift in the FMB Interaction Slope

Variable	Coefficient	t -stat	p -value
const	-0.000061	-0.41	0.684
Post-2020 indicator	-0.000034	-0.14	0.890

The Chow test finds no evidence that the PE–SAT interaction strengthened after December 2019 (the point estimate is in fact slightly negative and $|t| < 0.5$). Hypothesis 4 — the

paper’s sharpest structural prediction — is not supported. Figure 2 plots the 36-month rolling Δ_{Cross} : it oscillates around zero over the full sample with no visible level shift at the regime break.

8.6 Daily-Horizon Analysis

If the monthly aggregation were masking a short-lived signal, daily sorts would recover it. We therefore rebuild the entire design at daily frequency: three SAT variants (raw daily, 5-day rolling, 10-day rolling) are computed for every stock-day, stocks are sorted daily into 5×5 PE/SAT cells, and portfolio returns are measured on day $d+1$ (with forward returns masked across data gaps longer than one calendar week). Daily Fama–MacBeth regressions mirror (8) with Newey–West lags set to 22 ($\approx$ one trading month).

Table 7: Daily Fama–MacBeth Regressions: Three SAT Variants (Full Sample)

Variable	SAT raw		SAT 5-day		SAT 10-day	
	$\hat{\lambda}$	t	$\hat{\lambda}$	t	$\hat{\lambda}$	t
const	0.000439	2.03**	0.000646	2.81***	0.000715	3.17***
PEQ	-0.000002	-0.07	-0.000023	-0.64	-0.000047	-1.35
SATQ	0.000066	2.76***	-0.000002	-0.06	-0.000026	-0.98
PEQ×SATQ	-0.000011	-1.67*	-0.000004	-0.64	0.000004	0.65

Notes: Each trading day provides one cross-section; the dependent variable is the day- $d+1$ excess return, winsorised at 0.5%/99.5% within each day. Newey–West (22 lags) on the time series of daily slopes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Daily Δ_{Cross} by SAT Variant

SAT variant	Δ_{Cross} (%/day)	t (NW22)	N days
Raw daily	-0.0380	-1.81*	3936
5-day rolling	-0.0089	-0.40	3932
10-day rolling	0.0195	0.95	3922

The daily design covers 4,268 trading days, 600 stocks, and 2.26 million stock-days. It recovers no signal either. The daily Δ_{Cross} is insignificant for all three SAT variants — and for the raw variant its point estimate has the *wrong* sign. In the daily Fama–MacBeth regressions the PE slope and the PE×SAT interaction are null at every horizon. The single marginally significant coefficient — a SAT slope of 0.66 basis points per day per quintile rank in the raw variant ($t = 2.76$) — disappears completely in the 5-day and 10-day smoothed variants ($t = -0.06$ and -0.98). A one-day effect that vanishes under a

five-day moving average of the same signal is the signature of market-microstructure noise (bid–ask bounce and transient price pressure), not of a sentiment premium; at two-thirds of a basis point per day per rank it is also economically irrelevant after any spread cost. The daily evidence therefore reinforces, rather than qualifies, the monthly null.

8.7 Robustness

The null is robust. Equal-weighted portfolios produce $\Delta_{\text{Cross}} = 0.03\%$ /month; the small/mid-cap subsample (excluding the top 30% by market capitalisation, where retail effects should be strongest) produces $\Delta_{\text{Cross}} = 0.09\%$ /month; neither is significant. The corner long–short portfolio — long (PE Q5, SAT Q5), short (PE Q1, SAT Q1) — earns a raw 0.09% per month, three orders of magnitude below transaction-cost relevance for a strategy with monthly turnover of this magnitude.

9 Look-Ahead Diagnostics: How Signed-Volume Measures Manufacture Alpha

An earlier draft of this paper reported SAT spreads of 8.0–9.4% per month with Newey–West t -statistics between 19.9 and 29.3, and FF6 alphas from -4.11% to $+6.46\%$ per month, 22 of 25 significant. Those numbers were produced by a portfolio construction that paired the month- t sort with month- t returns. Because SAT is signed by the formation-month return (Remark 1), the sort variable and the “holding” return shared the same directional information: stocks land in SAT Q5 largely *because* their month- t returns were positive. Table 9 places the two sets of headline statistics side by side.

Table 9: Contaminated versus Corrected Headline Statistics

Statistic	Contaminated (sort t , return t)	Corrected (sort t , return $t+1$)
SAT spread range (%/month)	8.0 to 9.4	-0.15 to 0.24
Max $ t $ on SAT spread	29.3	0.65
FF6 alpha range (%/month)	-4.11 to +6.46	-0.19 to +0.67
Alphas with $ t > 2$	22 of 25	2 of 25
Δ_{Cross} (%/month)	+0.88 ($t = 1.54$)	-0.05 ($t = -0.10$)
FMB λ_{Int}	insignificant	insignificant

Notes: “Contaminated” pairs month- t sorts with month- t returns; “Corrected” holds the same sorted portfolios over month $t+1$. Same panel, same code base otherwise. The Fama–MacBeth row is unchanged because the FMB implementation always used forward returns.

Three features make this diagnostic worth reporting rather than silently fixing. First,

the magnitude: contamination produced t -statistics an order of magnitude above the [Harvey et al. \(2016\)](#) threshold, from a signal with zero genuine predictive content. Any signed-volume or signed-order-flow measure — order imbalance, signed turnover, signed dollar volume — carries the same hazard, and several literatures build on such measures. Second, the detectability: the contaminated results contained an internal inconsistency (portfolio sorts implying enormous premia while properly lagged FMB regressions found nothing), which is exactly the symptom replicators should look for. Third, the fix is structural, not judgemental: a single shared forward-return code path, plus automated tests in which the sorting variable drives only formation-month returns and any nonzero portfolio return fails the build.

10 RSF as a Complementary Factor: Companion Results

The null predictive result does not exhaust the asset-pricing content of retail sentiment. A variable can fail to *forecast* returns and still capture a distinct source of *contemporaneous* common variation — which is precisely what the Fama–French factors themselves are: SMB does not forecast; it is estimated contemporaneously on the right-hand side and strips size-related comovement out of test-asset returns.

In a companion paper we construct RSF as a Fama–French-style factor: an independent 2×3 sort on size (median split) and SAT (30th/70th percentiles), value-weighted, formation at t and held over $t+1$, with $\text{RSF} = \frac{1}{2}(\text{Small/High} + \text{Big/High}) - \frac{1}{2}(\text{Small/Low} + \text{Big/Low})$. Over 185 months the factor earns 0.12% per month (annualised Sharpe 0.29). The key result is the spanning regression: regressed on the FF5 + momentum factors, RSF has an alpha of 0.22% per month ($t = 2.24$, $p = 0.025$) with $R^2 = 0.03$ and no correlation with any FF factor exceeding 0.12 in absolute value. The FF factors do not span RSF: retail sentiment carries common return variation the standard model misses, even though it forecasts nothing. The companion paper develops the construction, the spanning tests, and the effect of adding RSF as a seventh right-hand-side factor.

11 Conclusion

European retail equity participation doubled between 2018 and 2022. This paper asked whether that shift left a predictive footprint in the cross-section of European equity returns, and the answer — under a strictly enforced sort-at- t / hold-at- $t+1$ convention — is no. SAT quintile spreads are economically negligible and statistically indistinguishable from zero; FF6 alphas show no sentiment gradient; Fama–MacBeth slopes are flat; the PE–SAT interaction is zero in both the full sample and around the 2020 regime break;

and daily-frequency variants of the design reach the same conclusion. Hypotheses 1–4 are not supported.

The paper’s second contribution is the anatomy of how the opposite conclusion was nearly published. Signed-volume sentiment measures embed the formation-period return sign by construction, so a one-month misalignment between sort and holding period does not merely bias results — it manufactures them at a scale ($t > 19$) that swamps every conventional robustness check. The contaminated results survived value/equal weighting, subsample splits, and FF6 risk adjustment, because the contamination was orthogonal to all of those. What exposed it was alignment-sensitive cross-validation: the Fama–MacBeth regressions, which used properly lagged returns, kept returning nulls that the portfolio sorts contradicted. We recommend that disagreement between sort-based and regression-based estimates of the same premium be treated as an alignment red flag, particularly for signed measures.

What survives is the reinterpretation. Retail sentiment, packaged as an FF-style factor, is not spanned by the Fama–French five factors plus momentum: it earns a significant spanning alpha with near-zero factor correlations. The constructive role of the RSF is therefore on the right-hand side of factor models — a control for sentiment-driven comovement — rather than in a forecasting rule. The companion paper develops this in full.



Figure 1: Mean monthly excess return by portfolio cell — value weighted (corrected alignment).

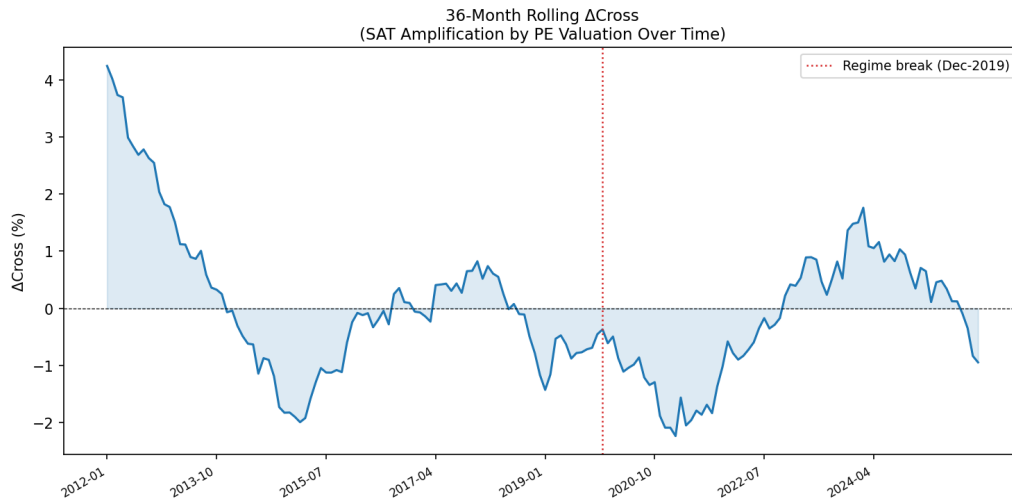


Figure 2: 36-month rolling Δ_{Cross} (corrected alignment).

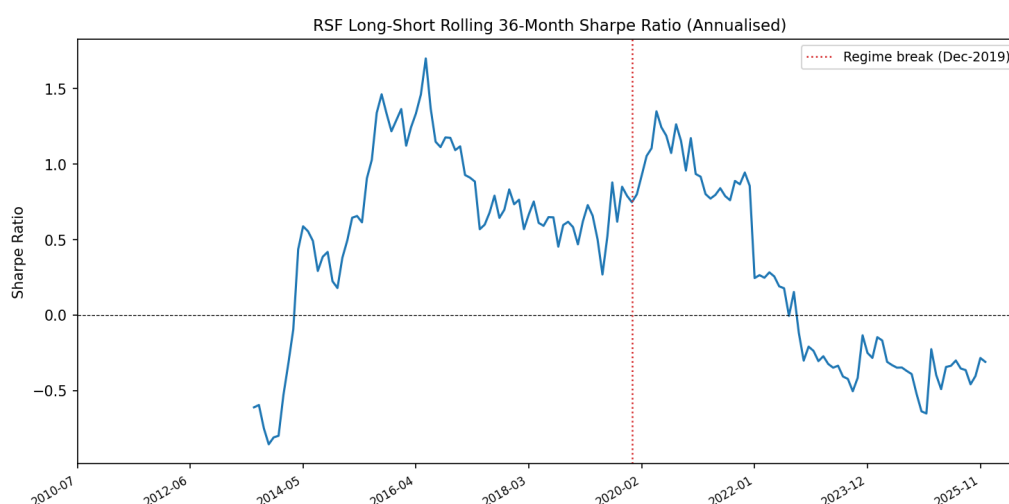


Figure 3: RSF corner long-short: rolling 36-month annualised Sharpe ratio (corrected alignment).

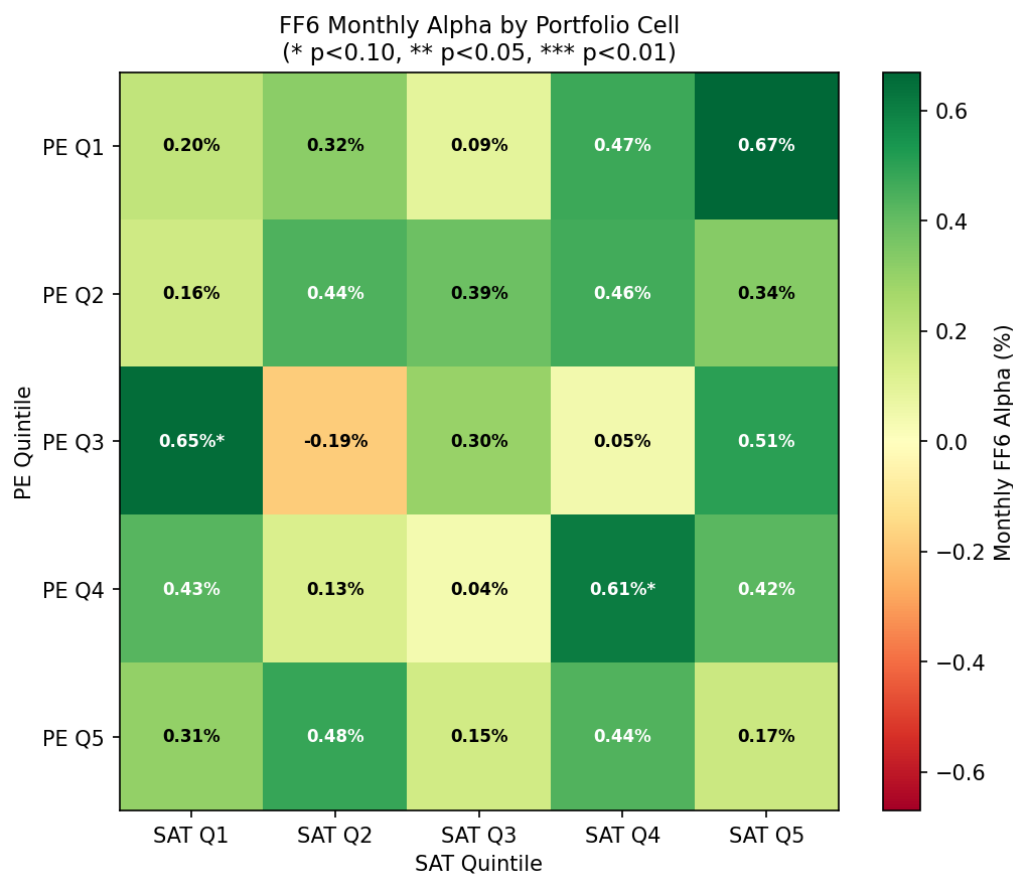


Figure 4: FF6 monthly alpha by portfolio cell with significance stars (corrected alignment).

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